

ΣΥΣΤΗΜΑΤΑ ΑΝΑΜΟΝΗΣ (Queuing Systems)

Forecasting – Model Fitting



Book from where to read:

Performance Evaluation of Computer and Communication Systems, **Jean-Yves Le Boudec**, EPFL Press, Distributed by
CRC Press

Contents

1. What is forecasting ?
2. Model Fitting with Least Squares
3. Linear Regression
4. Avoiding Overfitting
5. Use of Bootstrap

1. What is forecasting?

- ✓ Assume you have been able to define the *nature* of the load for your study
- It remains to have an idea about its *intensity*

It is impossible to forecast without error

The good engineer should

- Forecast *what can be forecasted*
- Give *uncertainty* intervals

The rest is outside our control

Examples that forecasting is needed

- Capacity Planning e.g., queue capacities so that blocking probability is small
- Optimal use of resources, e.g., determine service rates

What is forecasting?

Forecasting = finding conditional distribution of future given past

Assume we observe some data $Y_1, Y_2, \dots, Y_t, \dots$

We have observed $Y_1, Y_2, \dots, Y_t$ and want to forecast $Y_{t+\ell}$

Full Forecast: Conditional distribution of $Y_{t+\ell}$ given $Y_1, Y_2, \dots, Y_t$

Point Forecast: $\hat{Y}_t(\ell) = E(Y_{t+\ell} | Y_1, \dots, Y_t)$

Prediction interval [A;B] at level $(1 - \epsilon)\%$:

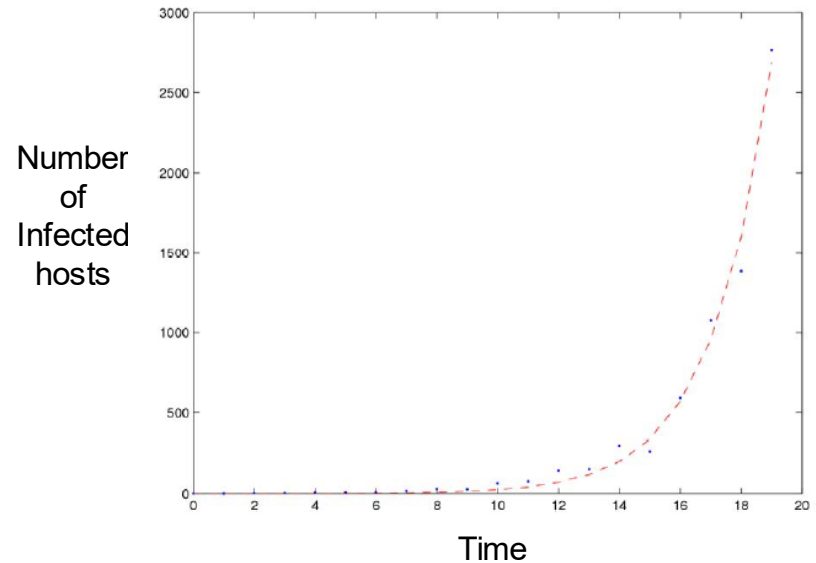
$$P(A \leq Y_{t+\ell} \leq B | Y_1, Y_2, \dots, Y_t) > 1 - \epsilon$$

2. Some Background on Model Fitting

We would like to capture the growth of infected hosts
(explanatory model)

An exponential model seems appropriate

How can we fit the model, in particular, what is the value of its parameters?



$$Y(t) = ae^{\alpha t}$$

What is model fitting?

To find an explanatory model for the data.

Given some data Y_i and a parametric model $f_i(\theta)$ we assume that:

$$Y_i = f_i(\theta) + \text{noise}$$

The model fitting problem is to estimate the unknown parameter θ

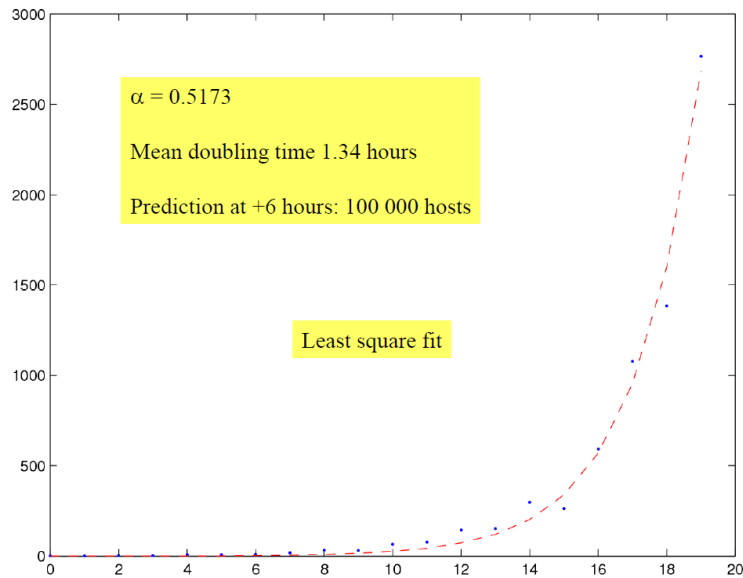
For example: the model is $Y_i = a e^{\alpha t_i}$

or
$$\log Y_i = \log a + \alpha t_i$$

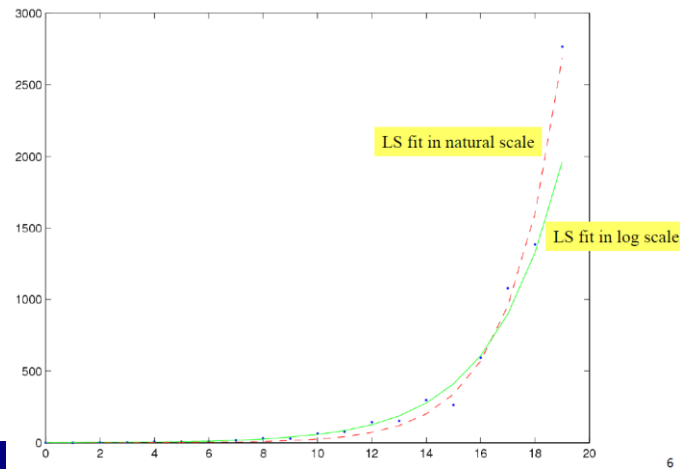
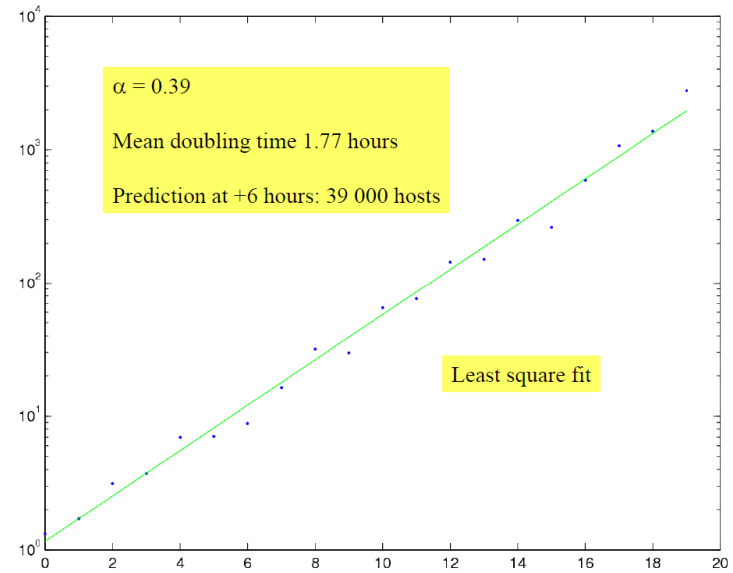
Both models are equivalent with parameter $\theta = (a, \alpha)$

Model Fitting Examples

$$Y_i = a e^{\alpha t_i}$$



$$\log Y_i = \log a + \alpha t_i$$



What is model fitting?

Signal processing interpretation

Given a model $Y_i = f_i(\theta) + \text{noise}$ and a **score function**, find the parameter value θ that minimizes the score function.

Classical choices for the score

Least Square: $\text{score} = \sum_i (\text{noise})^2$

Weighted Least Square: $\text{score} = \sum_i (w_i \times \text{noise})^2$

ℓ^1 score: $\text{score} = \sum_i |\text{noise}|$

Weighted ℓ^1 score: $\text{score} = \sum_i w_i \times |\text{noise}|$

Which one should we use? -> answer by using the statistical interpretation

What is model fitting?

The statistical interpretation

The data Y_i is the output of a statistical system, a simulator, which has a parameter θ that we do not know and want to estimate.

We observe only the output of the simulator.

We need to model the simulator including noise. This forces us to make some assumption about the noise term. For example, one of the following models can be considered:

1. $Y_i = a e_i^{\alpha t} + \epsilon_i$, ϵ_i i.i.d. $\sim N(0, \sigma^2)$ $\longrightarrow$ Error has same magnitude at all points
2. $Y_i = a e_i^{\alpha t} (1 + \epsilon_i)$, ϵ_i i.i.d. $\sim N(0, \sigma^2)$ $\longrightarrow$ Error increases proportionally to the value
3. $\log Y_i = \log a + \alpha t_i + \epsilon_i$, ϵ_i i.i.d. $\sim N(0, \sigma^2)$ $\longrightarrow$ Error has same magnitude at all points

These models are not equivalent.

For each of those models the parameter is now $\theta = (a, \alpha, \sigma)$.

Estimating a statistical model: maximum likelihood

Principle: find θ that maximizes the pdf $f_Y(y|\theta)$, also called likelihood where y is the data $\rightarrow$ *main intuition is that the data I have seen are more likely than those I have not.*

This estimation method is Robust (with high number of data does not depend on scale, i.e., it is the same for equivalent models, e.g., log or natural scales) and Consistent (in the long run with high amount of data you get the true value) .

Example: model 1 : $Y_i = ae^{\alpha t_i} + \epsilon_i, \quad \epsilon_i \text{ iid } \sim N(0, \sigma^2)$

$$f_Y(y|a, \alpha, \sigma) = \frac{1}{(\sqrt{2\pi}\sigma)^I} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^I (y_i - ae^{\alpha t_i})^2} \quad (1)$$

the problem is to find (a, α, σ) that maximizes (1)

And this gives a simple result for the model 1 that is called homoscedasticity
 $\rightarrow$ (next slide)

Least Square Score= Gaussian i.i.d. noise

Assume model (homoscedasticity)

$$Y_i = f_i(\vec{\beta}) + \epsilon_i \text{ for } i = 1, \dots, I \text{ with } \epsilon_i \text{ iid } \sim N_{0, \sigma^2}$$

THEOREM 3.1.1 (Least Squares). *For the model in Eq.(3.5),*

1. *the maximum likelihood estimator of the parameter $(\vec{\beta}, \sigma)$ is given by:*

$$(a) \quad \hat{\beta} = \arg \min_{\vec{\beta}} \sum_i \left(y_i - f_i(\vec{\beta}) \right)^2$$

$$(b) \quad \hat{\sigma}^2 = \frac{1}{I} \sum_i \left(y_i - f_i(\hat{\beta}) \right)^2 \rightarrow \text{Average square error for a value of beta}$$

i.e. the signal processing method with Least Squares score is the same as the statistical method with iid gaussian noise

Least Square Score Example

Example: model 1 : $Y_i = ae^{\alpha t_i} + \epsilon_i$, ϵ_i iid $\sim N(0, \sigma^2)$

$$f_Y(y|a, \alpha, \sigma) = \frac{1}{(\sqrt{2\pi}\sigma)^I} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^I (y_i - ae^{\alpha t_i})^2} \quad (1)$$

the problem is to find (a, α, σ) that maximizes (1)

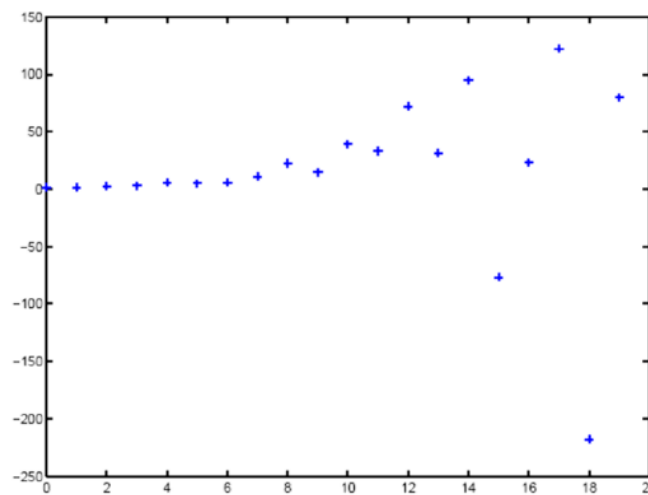
Thm 3.3.1 $\Rightarrow$

$$(\hat{a}, \hat{\alpha}) = \operatorname{argmin}_{a, \alpha} \sum_{i=1}^I (y_i - ae^{\alpha t_i})^2$$

$$\hat{\sigma}^2 = \frac{1}{I} \sum_i^I (y_i - \hat{a} e^{\hat{\alpha} t_i})^2$$

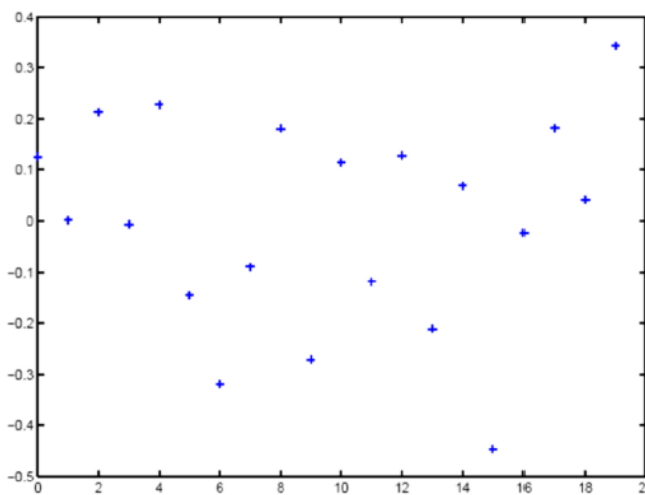
Validate the Fitting Method via the Residuals

Residuals = estimate of noise terms (after estimation of parameter)
They should be consistent with the model considered, i.e., $N(0, \sigma^2)$



Model 1:

$$Y_i = ae^{\alpha t_i} + \epsilon_i, \epsilon_i \sim iid N(0, \sigma^2)$$



Model 3:

$$\log Y_i = \log a + \alpha t_i + \epsilon_i, \epsilon_i \sim iid N(0, \sigma^2)$$

Observations: Model 1 appears not to fit because residuals have a variance that grows with time. Model 3 seems OK; we should prefer it for fitting this data set.

LS score

Given a model to be fitted

$$Y_i = f_i(\theta) + \text{noise}$$

we have the equivalences:

Score	Statistical model
Least Square: score = $\sum_i (\text{noise}_i)^2$	noise $\sim \text{iid } N(0, \sigma^2)$
Weighted Least Square: score = $\sum_i (w_i \times \text{noise}_i)^2$	noise _i $\sim \text{iid } N\left(0, \frac{\sigma^2}{w_i^2}\right)$

Why need of WLS: It is rare that the error produced is the same magnitude for all observations, e.g., consider observing an infection at an early stage and later after having spread

i.e., WLS gives a weight inversely proportional to the variance of the noise

Questions

- A. A1 B2 C3 1. $Y_i = ae^{\alpha t_i} + \epsilon_i, \quad \epsilon_i \text{ iid } \sim N(0, \sigma^2)$
B. A1 B3 C2 2. $Y_i = ae^{\alpha t_i}(1 + \epsilon_i), \quad \epsilon_i \text{ iid } \sim N(0, \sigma^2)$
C. A2 B1 C3 3. $\log Y_i = \log a + \alpha t_i + \epsilon_i, \quad \epsilon_i \text{ iid } \sim N(0, \sigma^2)$
D. A2 B3 C1
E. A3 B1 C2 A: $\sum_i^I (Y_i - ae^{\alpha t_i})^2$ B: $\sum_i^I \left(\frac{Y_i - ae^{\alpha t_i}}{ae^{\alpha t_i}} \right)^2$
F. A3 B2 C1 C: $\sum_i^I (\log Y_i - \log a - \alpha t_i)^2$
G. I don't know

Compare models 2 and 3

1. $Y_i = ae^{\alpha t_i} + \epsilon_i, \quad \epsilon_i \text{ iid } \sim N(0, \sigma^2)$
2. $Y_i = ae^{\alpha t_i}(1 + \epsilon_i), \quad \epsilon_i \text{ iid } \sim N(0, \sigma^2)$
3. $\log Y_i = \log a + \alpha t_i + \epsilon_i, \quad \epsilon_i \text{ iid } \sim N(0, \sigma^2)$

Assume model 2 and take log:

$$\begin{aligned}\log Y_i &= \log a + \alpha t_i + \log(1 + \epsilon_i) \\ &\approx \log a + \alpha t_i + \epsilon_i\end{aligned}$$

So we expect models 2 and 3 to be approximately equivalent.

3. Linear Regression

Model 3 is a special case of linear regression.

Definition: A linear regression model (with least squares) is a model of the form

$$Y_i = f_i(\beta) + \epsilon_i, \epsilon_i \text{ i.i.d. } \sim N(0, \sigma^2)$$

where the function $f_i(\beta)$ is linear with respect to β .

DEFINITION 3.1 (Linear Regression Model).

$$Y_i = (X\vec{\beta})_i + \epsilon_i \text{ for } i = 1, \dots, I \text{ with } \epsilon_i \text{ iid } \sim N_{0, \sigma^2} \quad (3.8)$$

where the unknown parameter $\vec{\beta}$ is in $\mathbb{R}^p$ and X is a $I \times p$ matrix. The matrix X supposed to be known exactly in advance. We also assume that

H X has rank p

For such models there exist closed form solutions in matrix form for the maximum likelihood estimation solution, including exact confidence intervals. $\rightarrow$ next slide

Closed Forms for Linear Regression

THEOREM 3.3 (Linear Regression). Consider the model in Definition 3.1; let $\vec{y}$ be the $I \times 1$ column vector of the data.

1. The $p \times p$ matrix $(X^T X)$ is invertible
2. (Estimation) The maximum likelihood estimator of $\vec{\beta}$ is $\hat{\beta} = K\vec{y}$ with $K = (X^T X)^{-1} X^T$
3. (Standardized Residuals) Define the i th residual as $e_i = (\vec{y} - X\hat{\beta})_i$. The residuals are zero-mean gaussian but are correlated, with covariance matrix $\sigma^2(I_d - H)$, where $H = X(X^T X)^{-1} X^T$.

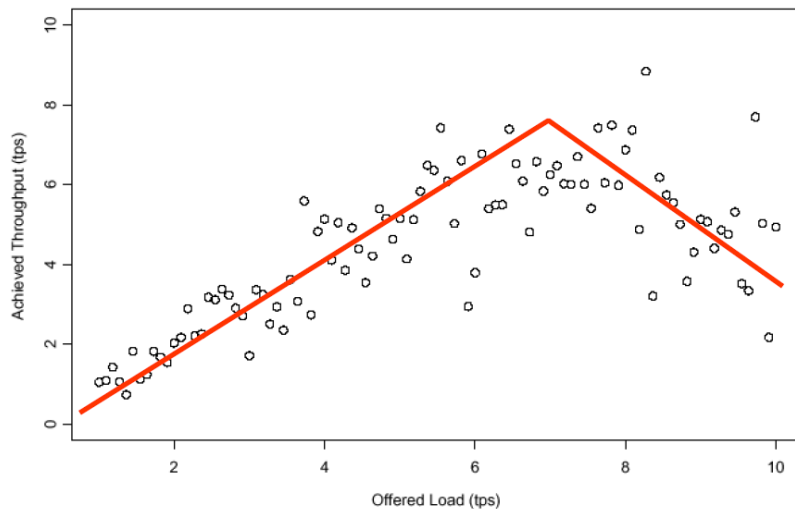
Let $s^2 = \frac{1}{I-p} \|e\|^2 = \frac{1}{I-p} \sum_i e_i^2$ (rescaled sum of squared residuals). s^2 is an unbiased estimator of σ^2 .

The standardized residuals defined by $r_i := \frac{e_i}{s\sqrt{1-H_{i,i}}}$ have unit variance and $r_i \sim t_{I-p-1}$. This can be used to test the model by checking that r_i are approximately normal with unit variance.

4. (Confidence Intervals) Let $G = (X^T X)^{-1} = K K^T$; the distribution of $\hat{\beta}$ is gaussian with mean $\vec{\beta}$ and covariance matrix $\sigma^2 G$, and $\hat{\beta}$ is independent of e .

In particular, assume we want a confidence interval for a (non-random) linear combination of the parameters $\gamma = \sum_{j=1}^p u_j \beta_j$; $\hat{\gamma} = \sum_j u_j \hat{\beta}_j$ is our estimator of γ . Let $g = \sum_{j,k} u_j G_{j,k} u_k = \sum_k \left(\sum_j u_j K_{j,k} \right)^2$ (g is called the **variance bias**). Then $\frac{\hat{\gamma} - \gamma}{\sqrt{gs}} \sim t_{I-p}$. This can be used to obtain a confidence interval for γ .

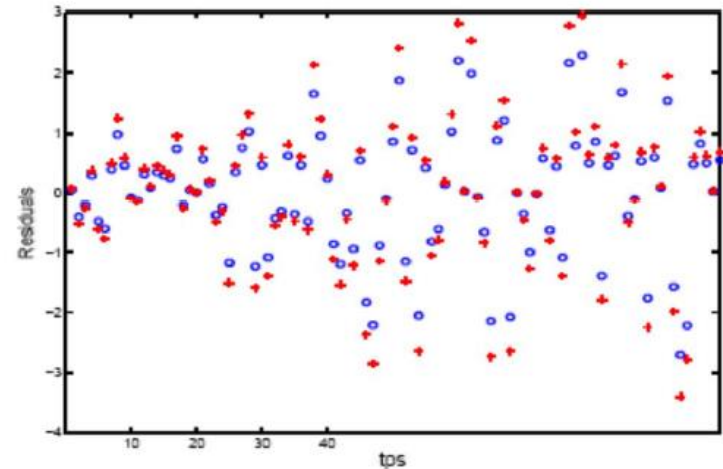
Example – Piece wise linear



$$Y_i = (a + bx_i)1_{x_i \leq \xi} + (c + dx_i)1_{x_i > \xi} + \epsilon_i, \epsilon_i \sim \text{iid } N(0, \sigma^2)$$

with $a + b\xi = c + d\xi$

This is a linear regression only if ξ is known. Then the dependence on (a, b, c, d) is linear and the noise is iid gaussian.
If ξ needs to be fitted as well then it is non-linear.



Residuals are slightly smaller for smaller load values.

Regression Models

Simple, often used

Based on a model fitted over the past, assumed to hold in the future

$$Y_t = \sum_{j=1}^p \beta_j f_j(t) + \epsilon_t$$



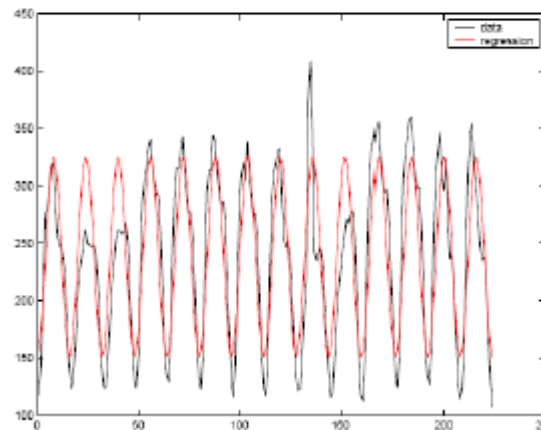
Known, non random functions,
not necessarily linear w.r.t. t

Example:

$$Y_t = 238.2475 - 871876 \cos\left(\frac{\pi}{8}t\right) - 4,2961 \sin\left(\frac{\pi}{8}t\right) + \epsilon_t$$

with $\epsilon_t \sim iid N(0, \sigma^2)$,

and $\sigma = 38.2667$



- Aim is to resell some capacity and want to predict the load in the future.

Regression Models

Prediction

We have obtained the model

$$Y_t = 238.2475 - 871876 \cos\left(\frac{\pi}{8}t\right) - 4.2961 \sin\left(\frac{\pi}{8}t\right) + \epsilon_t$$

with $\epsilon_t \sim iid N(0, \sigma^2), \sigma = 38.2667$

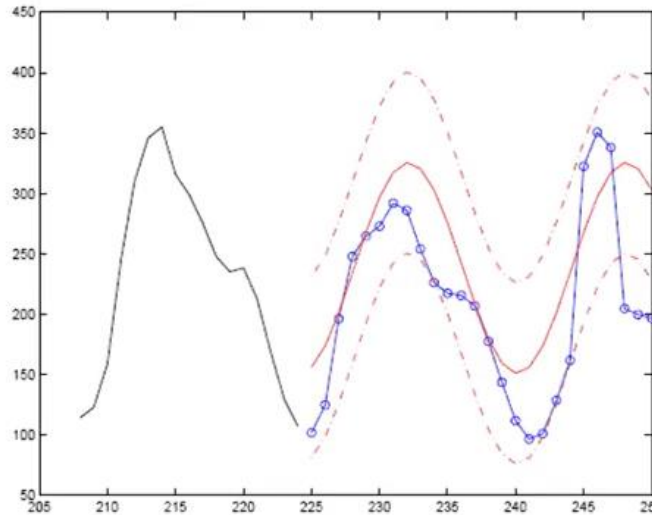
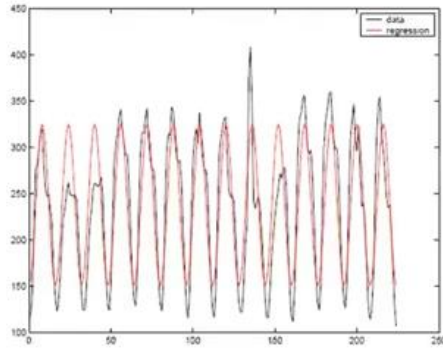
Thus $Y_{t+\ell} = 238.2475 - 871876 \cos\left(\frac{\pi}{8}(t + \ell)\right) - 4.2961 \sin\left(\frac{\pi}{8}(t + \ell)\right) + \epsilon_{t+\ell}$
with $\epsilon_{t+\ell} \sim N(0, \sigma^2), \sigma = 38.2667$ and $\epsilon_{t+\ell}$ independent of $Y_{1:t}$ (because $\epsilon_{t+\ell}$ is independent of $\epsilon_{1:t}$)

Point prediction:

$$\begin{aligned}\hat{Y}_t(\ell) &= \mathbb{E}(Y_{t+\ell} | Y_{1:t}) \\ &= 238.2475 - 871876 \cos\left(\frac{\pi}{8}(t + \ell)\right) - 4.2961 \sin\left(\frac{\pi}{8}(t + \ell)\right)\end{aligned}$$

because $\mathbb{E}(\epsilon_{t+\ell} | Y_{1:t}) = \mathbb{E}(\epsilon_{t+\ell}) = 0$

Regression Models



We have obtained

$$Y_{t+\ell} = 238.2475 - 871876 \cos\left(\frac{\pi}{8}(t+\ell)\right) - 4.2961 \sin\left(\frac{\pi}{8}(t+\ell)\right) + \epsilon_{t+\ell}$$

$$Y_{t+\ell} = \hat{Y}_t(\ell) + \epsilon_{t+\ell}$$

$$\mathbb{P}(\epsilon_{t+\ell} \in [-1.96\sigma ; 1.96\sigma] | Y_{1:t}) = \mathbb{P}(\epsilon_{t+\ell} \in [-1.96\sigma ; 1.96\sigma]) = 0.95$$

Thus a 95% **prediction interval** is $\hat{Y}_t(\ell) \pm 1.96\sigma$

Exact Formulas for Linear Regression with LS

THEOREM 5.1. Consider a linear regression model as in Eq.(5.1) with p degrees of freedom for $\vec{\beta}$. Assume that we have observed the data at n time points $t_1, \dots, t_n$, and that we fit the model to these n observations using Theorem 3.3. Assume that the model is regular, i.e. the matrix X defined by $X_{i,j} = f_j(t_i)$, $i = 1, \dots, n$, $j = 1, \dots, p$ has full rank. Let $\hat{\beta}_j$ be the estimator of β_j and s^2 the estimator of the variance, as in Theorem 3.3.

1. The point prediction at time $t_n + \ell$ is $\hat{Y}_{t_n}(\ell) = \sum_{j=1}^p \hat{\beta}_j f_j(t_n + \ell)$
2. An exact prediction interval at level $1 - \alpha$ is

$$\hat{Y}_{t_n}(\ell) \pm \xi \sqrt{1 + g} s \quad (5.3)$$

with

$$g = \sum_{j=1}^p \sum_{k=1}^p f_j(t_n + \ell) G_{j,k} f_k(t_n + \ell)$$

where $G = (X^T X)^{-1}$ and ξ is the $(1 - \frac{\alpha}{2})$ quantile of the student distribution with $n - p$ degrees of freedom, or, for large n , of the standard normal distribution.

3. An approximate prediction interval that ignores estimation uncertainty is

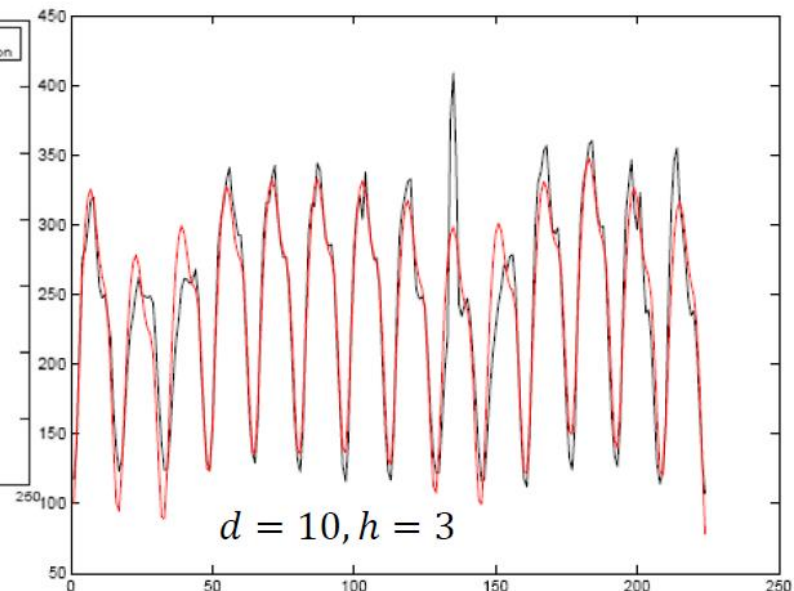
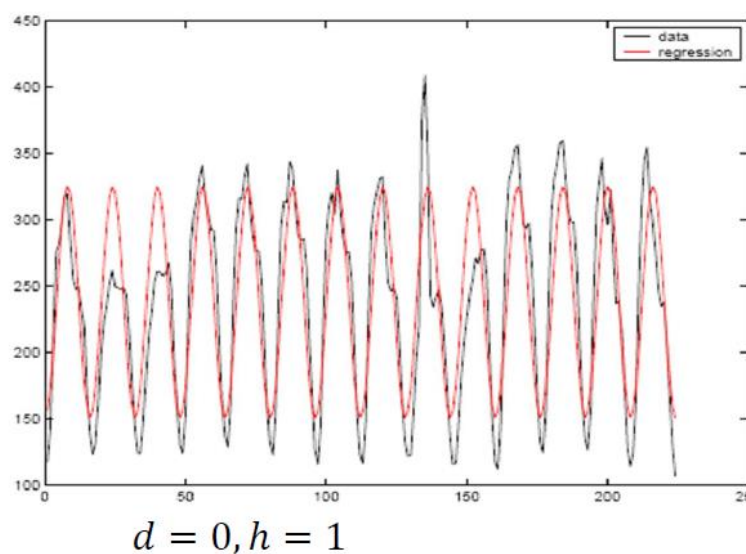
$$\hat{Y}_{t_n}(\ell) \pm \eta s \quad (5.4)$$

where η is the $1 - \alpha$ quantile of the standard normal distribution.

4. The Overfitting Problem

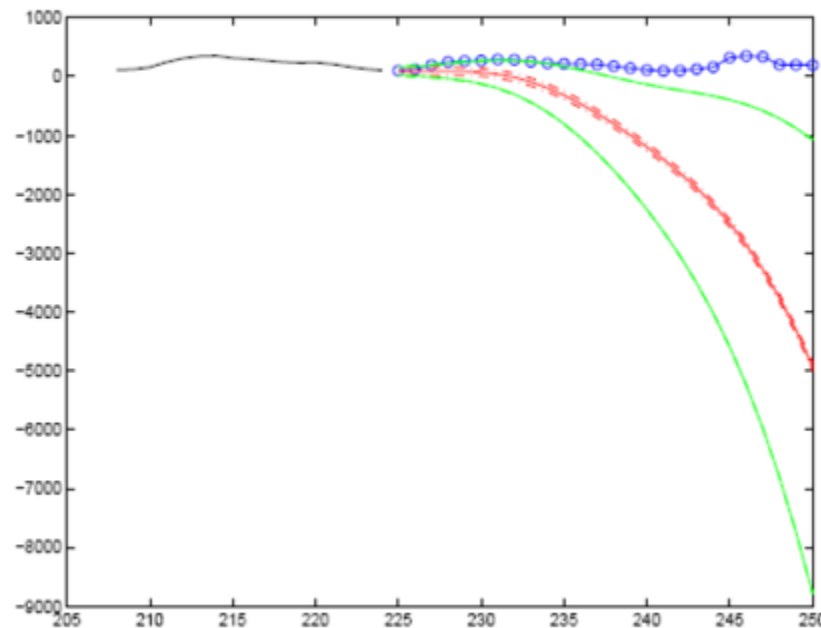
Assume that we aim to improve the model by adding more parameters e.g., a polynomial term and more harmonics with the hope of improving the fit and thus the prediction. The new model has the form:

$$Y_t = \sum_{i=0}^d a_i t^i + \sum_{j=1}^h \left(b_j \cos \frac{j\pi t}{8} + c_j \sin \frac{j\pi t}{8} \right)$$



The Overfitting Problem

Figure 6.4 shows the resulting fit for a polynomial of degree $d = 10$ and with $h - 1 = 2$ harmonics. The fit is better ($\sigma = 25.4375$ instead of 38.2667), however, the prediction power is ridiculous. This

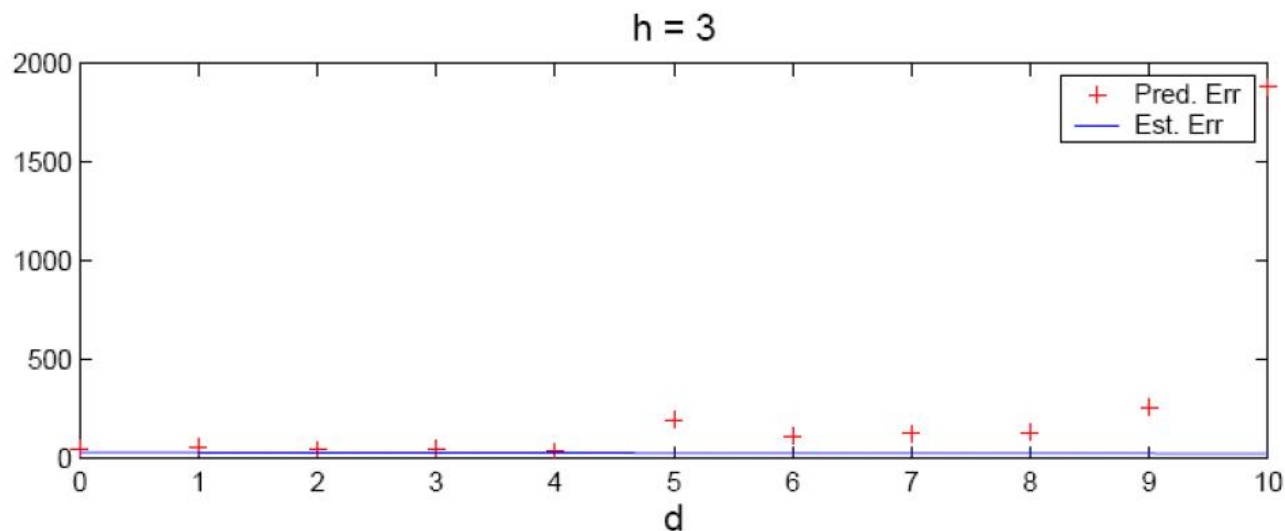


This is the overfitting problem: a better fit is not the best predictor, a model can fit the data but it is unable to model it;

How to solve the Overfitting Problem with cross-validation

Method 1: reserve some data for testing

The idea is to reserve a small fraction of the data set to test the model prediction. Consider for example Figure 6.5. We fitted the model in Eq.(6.5) with $h - 1 = 2$ harmonics and a polynomial of degree $d = 0$ to 10. The prediction error is defined here as the mean square error between the true values of the data at $t=225$ to 250 and the point predictions given by Theorem . The estimation error is the estimator s of σ . The smallest prediction error is for $d = 4$. The fitting error decreases with d , whereas the prediction error is minimal for $d = 4$. This method is quite



How to solve the Overfitting Problem with Information Criteria (1)

We use all the data.

Main idea: add a penalty to the score function to help us decide the optimal number of parameters.

The log-likelihood function $\log f_Y(y)$ was used to derive a score function to be minimized for model fitting.

For example, for a linear homoscedastic model with n data points:

$$\text{score} = -\log f_Y(y) = n \log \hat{\sigma} + \text{constant},$$

with $\hat{\sigma}$ the maximum likelihood estimator of σ

To avoid overfitting $\rightarrow$ add a penalty term to the score

Akaike's Information Criteria: $AIC = -2 \log f_Y(y) + 2k$, where k is the number of free parameters to be fitted

For instance, for a linear homoscedastic model with *parameter* $\beta \in R^p$ we have $k = p + 1$ and

$$AIC = 2n \log \hat{\sigma} + 2p + \text{constant}$$

AIC can be interpreted as the amount of information required to describe a new hypothetical sample when we estimate the model from one sample.

How to solve the Overfitting Problem with Information Criteria (2)

Other information criteria are also used. They are defined empirically.

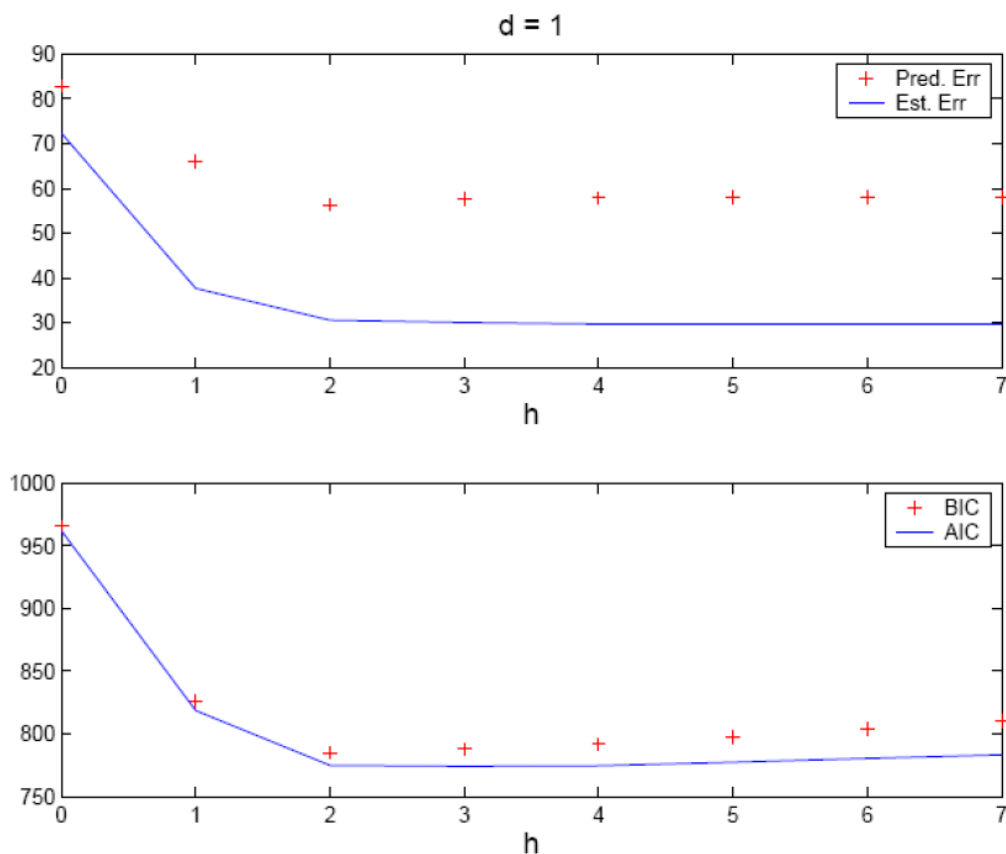
For example: the Bayesian Information Criterion (BIC) is defined for linear regression models

$$BIC = 2n \log \hat{\sigma} + 2p \log n + \text{constant}$$

- ✓ BIC gives more weight than AIC to model dimension p when the sample size n is large.

Overfitting Example (1)

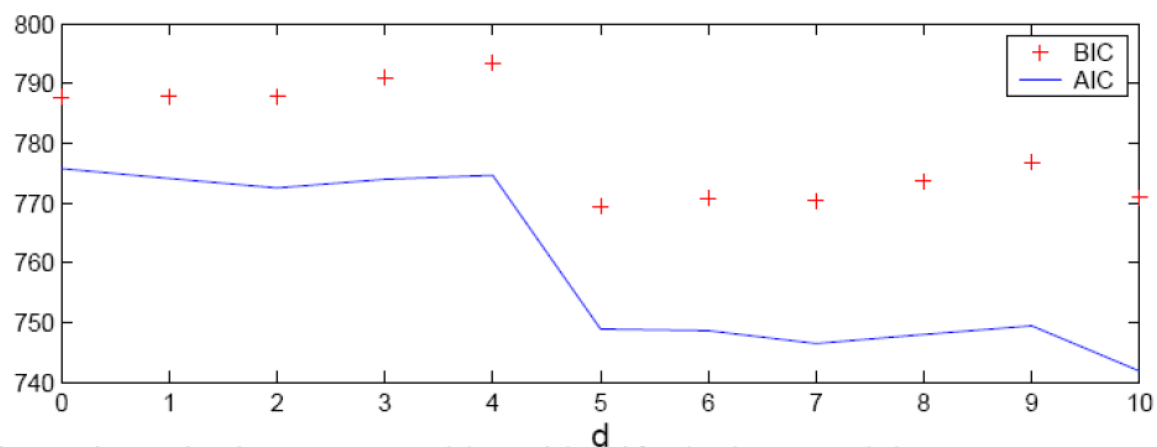
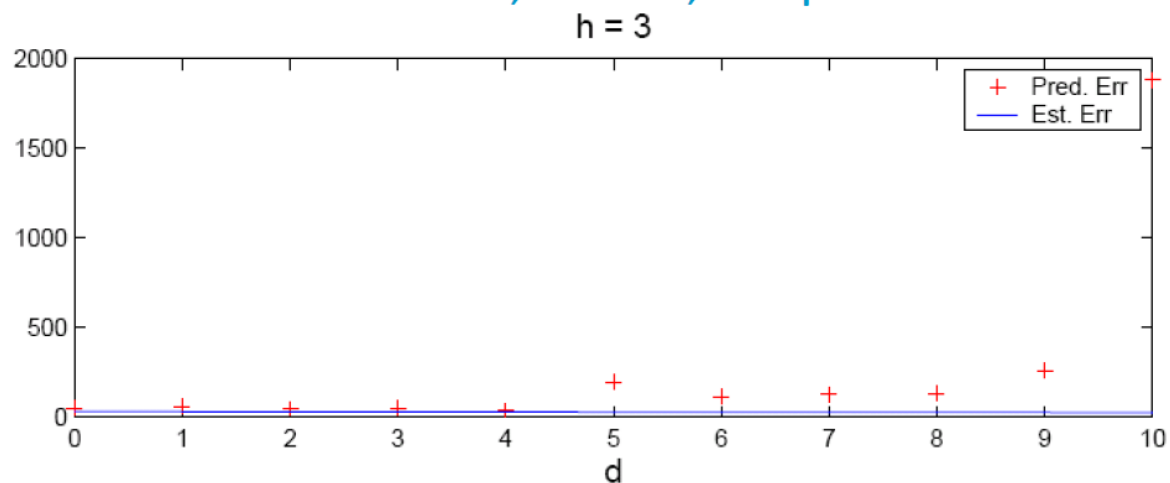
Best Model for Internet Data, $d = 1$, h up to 10



Information criteria are able to identify the best model

Overfitting Example (2)

Best Model for Internet Data, $h = 3$, d up to 10



Information criteria are not able to identify the best model;
the polynomial models are not a good class of models

5. Use of Bootstrap

Assume that we have a prediction model $Y_t = f_t(\beta) + \epsilon_t$

The estimation of β is done assuming some distribution for ϵ_t

Assume this distribution is only approximately known; we can improve the prediction intervals if we use a better approximation of this distribution.

For example, we can use the principle of the Bootstrap, i.e., estimate the distribution of ϵ_t by its empirical distribution.

Bootstrap Method

THEOREM 2.5 (General IID Case). Let $X_1, \dots, X_n, X_{n+1}$ be an iid sequence and assume that the common distribution has a density. Let $X_{(1)}^n, \dots, X_{(n)}^n$ be the order statistic of $X_1, \dots, X_n$. For $1 \leq j \leq k \leq n$:

$$\mathbb{P}(X_{(j)}^n \leq X_{n+1} \leq X_{(k)}^n) = \frac{k-j}{n+1} \quad (2.32)$$

thus for $\alpha \geq \frac{2}{n+1}$, $[X_{(\lfloor (n+1)\frac{\alpha}{2} \rfloor)}^n, X_{(\lceil (n+1)(1-\frac{\alpha}{2}) \rceil)}^n]$ is a prediction interval at level at least $\gamma = 1 - \alpha$.

Assume $Y_t = f_t(\beta) + \epsilon_t$ and apply theorem 2.5 to

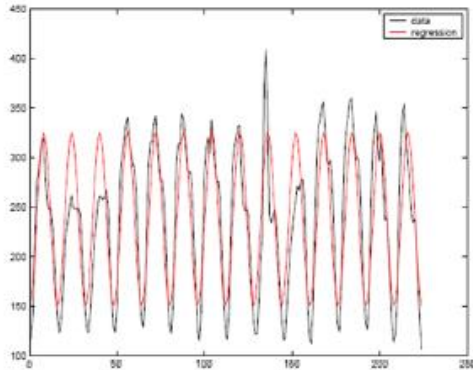
$$X_1 = \epsilon_1, \dots, X_n = \epsilon_t, X_{n+1} = \epsilon_{t+\ell}$$

This gives the algorithm:

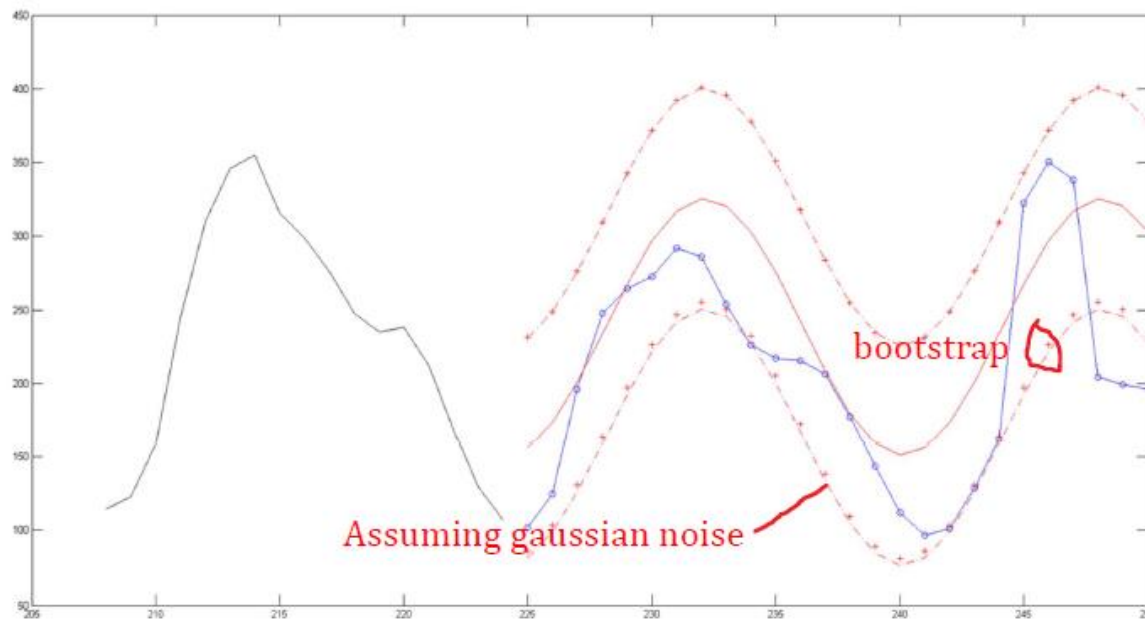
1. Estimate $\hat{\beta}$ by some method
2. Estimate residuals $e_t = Y_t - f_t(\hat{\beta})$
3. (Thm 2.5) $\eta = e_{\lfloor \frac{(t+1)\alpha}{2} \rfloor}, \xi = e_{\lceil (t+1)(1-\frac{\alpha}{2}) \rceil}$
(for $\alpha = 5\%, t = 100$: $\eta = e_{(2)}, \xi = e_{(99)}$)
4. Prediction interval for $Y_{t+\ell}$: $[f_{t+\ell}(\hat{\beta}) + \eta, f_{t+\ell}(\hat{\beta}) + \xi]$

Tractable method !!!

Bootstrap Examples



For this example, the bootstrap gives slightly smaller intervals than the ones based on gaussian noise



Prediction intervals with bootstrap are a bit optimistic, i.e., a bit smaller than the theoretical ones.